MACHINE LEARNING · FASTAPI · UVICORN

Mushroom Prediction

How we trained an edible-vs-poisonous classifier on 8,124 mushrooms and served it as an API with FastAPI on uvicorn.

github.com/ounmengheang/mushroom_prediction

OVERVIEW

Three stages, one repository

01

Dataset

8,124 labelled mushrooms described by 22 categorical features

`mushrooms.csv`

02

Train

Explore in a notebook, then a script trains and saves the model

`train_mushroom.py`

03

Serve

A FastAPI app on uvicorn exposes health, models and predict routes

`app/main.py`

01 · DATASET · MUSHROOMS.CSV

8,124 mushrooms, 22 clues each

8,124
rows · one mushroom each

22
categorical features (letter codes)

2
classes in the target column “class”

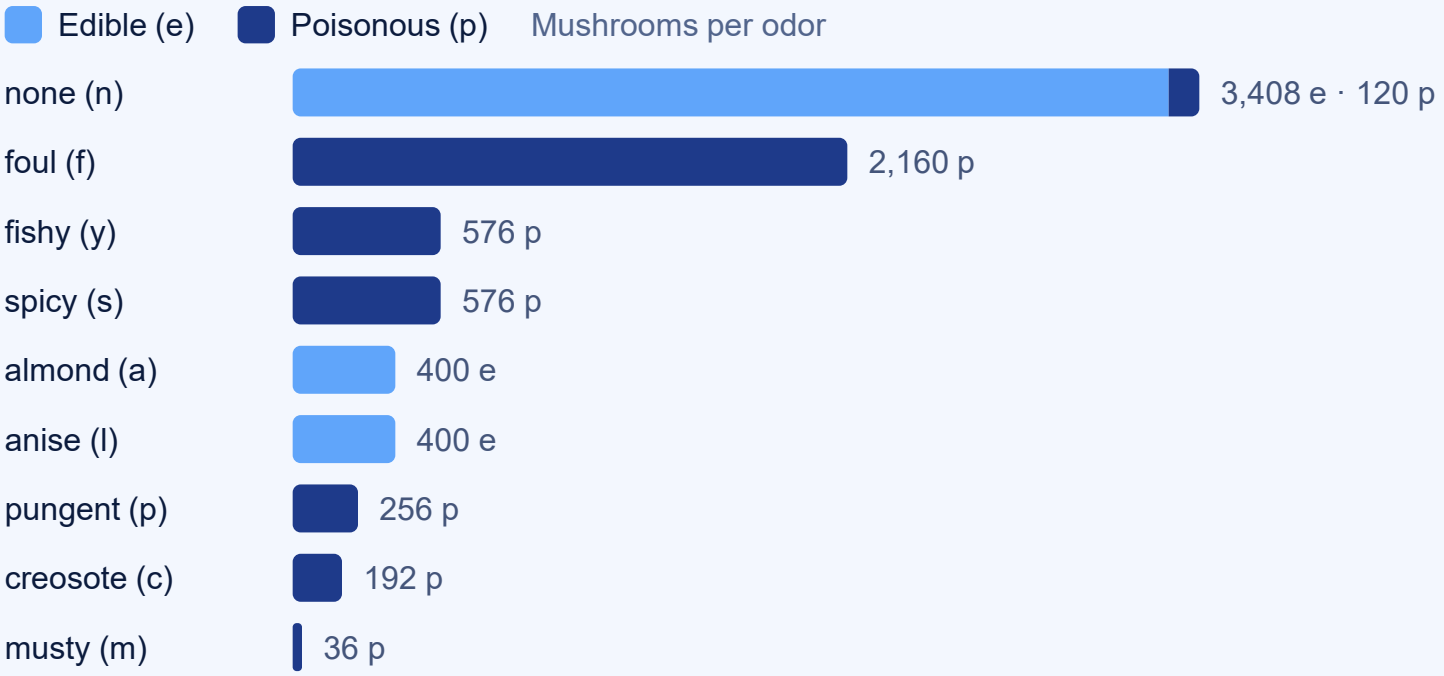
117
binary columns after one-hot encoding

CLASS BALANCE



01 · DATASET

Odor almost gives the answer away



8 of 9

odor values belong to a single class

Only “none” is mixed: 3,408 edible vs 120 poisonous. That is why even a simple linear model scores so high.

02 · TRAINING

The training pipeline, step by step

Load CSV
8,124 rows · 22 features + class

Stratified split
6,499 train / 1,625 test

One-hot encode
22 columns → 117 binary

Logistic regression
max_iter=1000

Score & save
3 metrics → mushroom.joblib

</> One command: **python train_mushroom.py**

02 · TRAINING

Near-perfect on 1,625 unseen mushrooms

99.94%

Accuracy on the test set

0.9994

Macro F1 (both classes equally weighted)

1.000

ROC AUC, poisonous as the positive class

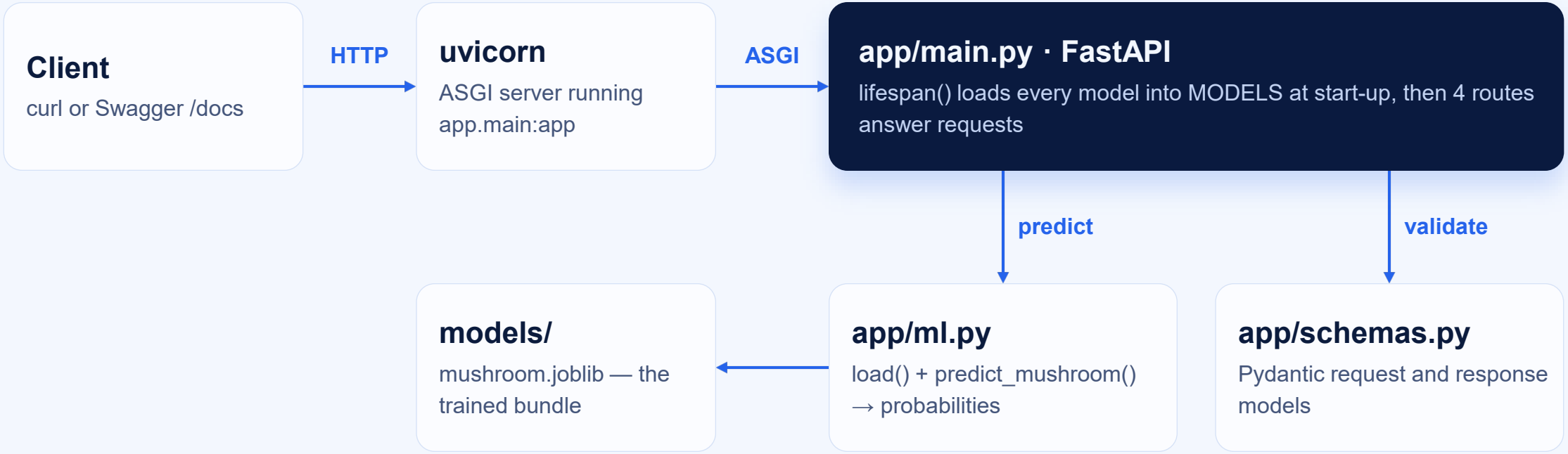
CONFUSION MATRIX · 1,625 TEST MUSHROOMS

	Predicted edible	Predicted poisonous
Actual edible	842	0
Actual poisonous	1	782

Only 1 of 1,625 test mushrooms is wrong (dashed).

03 · API

The request path, end to end



Start it with **uvicorn app.main:app --reload** (or **fastapi dev app/main.py**, which runs uvicorn for you), then open http://127.0.0.1:8000/docs

03 · API

Four routes, plus free docs

GET	/	Landing message that points to /docs and /health
GET	/health	Liveness → {"status": "ok", "models": {"mushroom": true}}
GET	/models	Model card: algorithm, trained_at and test metrics
POST	/mushroom/predict	22 features in → label, class_name, confidence, probabilities
GET	/docs	Swagger UI generated by FastAPI — try every route in the browser

03 · API

One request, one answer

POST /mushroom/predict

```
{
  "cap-shape": "x",
  "cap-surface": "s",
  "cap-color": "n",
  "bruises": "t",
  "odor": "p",
  "gill-attachment": "f",
  … 15 more features …
  "habitat": "u"
}
```

Same hyphenated names as the CSV columns — row 1 of mushrooms.csv



ml.py

200 OK

```
{
  "label": "p",
  "class_name": "poisonous",
  "confidence": 0.9859,
  "probabilities": {
    "e": 0.0141,
    "p": 0.9859
  }
}
```

Real response — correct, row 1 is labelled poisonous

WRAP-UP

Run it in three commands

1 Train
writes models/mushroom.joblib

```
$ python train_mushroom.py
```

2 Serve
ASGI server, auto-reload

```
$ uvicorn app.main:app --reload # or: fastapi dev app/main.py
```

3 Try
Swagger UI or curl

```
# open http://127.0.0.1:8000/docs, or:
$ curl -X POST http://127.0.0.1:8000/mushroom/predict \
    -H "Content-Type: application/json" -d @row.json
```

Needs: fastapi · uvicorn · scikit-learn · pandas · joblib

TAKEAWAYS

From CSV to a live prediction API

Data

Clean, balanced and fully categorical — odor alone nearly decides the class.

Model

One-hot + logistic regression: 99.94% accuracy and ROC AUC 1.000 on held-out data.

Service

FastAPI on uvicorn: 4 simple routes, validated input, instant predictions.

Thank you — questions?